

Ahmed Ikram | Software Engineering

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PROFILE

3rd-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Builds production-ready full-stack systems end-to-end - from React and React Native frontends through to secure Flask/FastAPI REST APIs, real-time Socket.IO layers, and AWS cloud infrastructure deployed via GitHub Actions CI/CD, with 1,446 automated tests enforcing 85% line coverage across every PR. Experienced leading a 7-person Agile team and independently architecting a 10-service Dockerised platform with JWT authentication, OAuth 2.0, three-tier RBAC, and OIDC-federated deployments using zero static credentials. Seeking a Software Engineering or Full-Stack post-graduation role to contribute across the entire stack in a team that ships production software.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

DevSync Project Management Platform with GitHub Integration | React, Tailwind, Python, Flask, PostgreSQL, Socket.io, GitHub OAuth, AWS, Docker, GitHub Actions, CI/CD | Jan 2025 - Apr 2025

- Built end-to-end: React, Tailwind frontend, Flask & Unicorn REST API over a 12-table indexed PostgreSQL schema, and a JWT-authenticated Socket.IO layer with project-scoped rooms - live task updates, notifications, and dashboard refreshes broadcast only to members viewing the affected project, with no page refresh required; GitHub activity fetched via GraphQL API to reduce latency when creating charts and reports
- Implemented three-tier RBAC (Developer / Team Leader / Admin) via role decorators on every route; JWT in HTTP-only cookies; GitHub OAuth 2.0 handled server-side with tokens never sent to the browser; bidirectional task ↔ issues/PR sync means closing a GitHub issue automatically updates the linked task in DevSync
- Production-like AWS infrastructure: ECS Fargate behind an ALB, RDS in a private subnet, and S3 & CloudFront for the React SPA - frontend deployment blocked until ECS health checks pass to prevent API/UI version mismatch; 1,446 automated tests at 85%/85% coverage hard-gate every PR before any AWS step runs, secrets injected at runtime via AWS Secrets Manager and per-task IAM roles scoped to least-privilege

ATM Log Aggregation, Analysis & Diagnostics Platform | Python, FastAPI, React, PostgreSQL, Kafka, Redis, nginx, Docker, AWS, MLflow | Jan 2025 - June 2025

- Led backend engineering on a 7-person Agile team delivering a diagnostics platform for NCR Atleos; independently architected an Apache Kafka event-driven pipeline, 3-layer ML detection engine, Agentic RAG assistant, and full 10-service Docker infrastructure, significantly extending the team-delivered project
- Architected a 10-service Dockerised system with health-check cascading startup (PostgreSQL → Kafka → services → frontend): FastAPI REST API (30 endpoints, 6 routers, JWT + Redis blacklist auth), 8 Redis patterns (rate limiting, distributed lock, Pub/Sub, DLQ, HyperLogLog, caching, dedup, JWT blacklist), and a React 19 frontend via multi-stage Docker build (~25MB nginx-alpine final image) with environment-parity proxy rewrites replacing the Vite dev server's '/api/*' routing in production nginx
- Validated the full system via 670 tests across 10 tiers (unit, integration, ML, RAG, Kafka, stress, security) with an isolated test PostgreSQL DB; all anomaly detection runs versioned in MLflow v3 on AWS (RDS + S3)

StockLens FinTech Spending & Investment App | React Native Expo, TypeScript, Node.js, PostgreSQL | Oct 2025 - Dec 2025

- Built a full-stack mobile app in React Native/TypeScript with secure biometric authentication, real-time Firebase synchronisation, and custom AES encryption for sensitive data at rest
- Integrated a Node.js OCR pipeline parsing physical receipts into structured JSON records, querying the Alpha Vantage API to map spending to stock tickers and compute historical CAGR - covered by 78 Jest tests across OCR fallbacks, API caching, cryptography roundtrips, auth flows and more
- Engineered an ML layer combining ARIMA-based time-series forecasting for per-category spending trends and Linear Regression for forward-looking portfolio projections, fed by historical CAGR computed from Alpha Vantage data - surfaced directly in the app UI

PROJECTS

- Rental Car Management System** | C++, CMake | Oct 2024 - Dec 2024
 - Designed an OOP system with a polymorphic vehicle hierarchy, a generic repository template pattern, and smart pointer memory management - zero raw pointer usage throughout
 - Implemented Levenshtein distance fuzzy search for vehicle and customer lookup, automated late fee and loyalty rewards logic, and an 8-scenario E2E test suite with file-based persistence
- Unix Version Control System** | Bash, Unix | Nov 2024 - Dec 2024
 - Built a Git-like VCS from scratch in pure Bash; contains file tracking, check-in/check-out with file locking, timestamped backup versioning, automatic diff generation on every check-in, and filterable activity logging with user attribution - zero external dependencies beyond native Unix utilities
 - Currently implementing branching and three-way merge support, an improved diff algorithm, and benchmarking performance against Git on representative workloads

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Bash
Frontend	React, React Native, HTML, CSS, Tailwind, Bootstrap, Vite, shadcn/ui, Chart.js
Backend	FastAPI, Flask, Node.js, REST APIs, WebSockets, GraphQL API, Apache Kafka, Redis, Socket.io, Gunicorn
Security	JWT, OAuth 2.0, RBAC, HTTP-only Cookies, OIDC Federation, AWS Secrets Manager
Databases	PostgreSQL, SQLite, MySQL, MongoDB, NoSQL
Cloud & DevOps	AWS (ECS Fargate, EC2, S3, RDS, IAM, SageMaker), Docker, nginx, GitHub Actions, CI/CD, Linux, Azure
Testing	Pytest, Jest, React Testing Library, Cypress, Integration, Stress & Security Testing

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	Feb 2025
<u>AWS Academy Graduate - Cloud Foundations</u>	Mar 2024
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	Oct 2024

EXPERIENCE

- Beaulys Mini Market, Dundee, Scotland**
Retail Assistant
September 2023 - Present
 - Troubleshoot and resolved Point-of-Sale and stock system errors quickly to minimise customer wait time
- Zulus Peri Peri Grill, Dundee, Scotland**
Front of House Worker
June 2021 - September 2023
 - Trained 5 new staff members, helping them get up to speed quickly during busy service periods

TRANSFERRABLE SKILLS

Agile & Iterative Delivery Comfortable owning work end-to-end within sprint cycles - scoping, building, testing, and shipping incrementally while adapting to changing requirements	Test-Driven Development & Code Quality Enforce quality gates via automated test suites (Pytest, Jest, Cypress) at 80–90% line coverage; treat test coverage as a non-negotiable, not an afterthought
System Design & Scalability Approach problems by reasoning about scalability, security, and maintainability from the start - not retrofitted after the fact	Technical Communication Explain API behaviour, architecture trade-offs, and system design decisions clearly to both technical peers and non-technical stakeholders
End-to-End Ownership Take full responsibility for delivery From requirements and user stories through schema design, API documentation, and architecture decisions through to cloud deployment and post-release monitoring against formal acceptance criteria	Handling Ambiguity Adept at breaking loosely defined problems into concrete technical tasks with clear acceptance criteria and measurable outcomes